

Final Project

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2024-08-10

Impacts of Urban Air Pollution on Respiratory Health

Introduction

The topic I will be addressing in this paper is the effects of urban air pollution on respiratory health. As urbanization continues, air pollution has become a major issue for people across the world. Air pollution has been linked to various negative health outcomes, including respiratory diseases like bronchitis, asthma, and lung cancer. Trying to understand the relationship between poor air quality and health outcomes is crucial for developing effective policies and interventions, and we can use the large datasets available to us to further understand this relationship.

Research Questions

How do the levels of air pollution vary across different cities?

What are the main pollutants contributing to health issues?

Does air pollution correlate with the rates of respiratory disease?

Can we predict respiratory health issues with air pollution data?

How effective are current air quality regulations?

Are there seasonal or weather-related variations in air pollution levels?

Approach

For this project I will first find datasets on air quality, air pollution, and respiratory health. I can then use this data to identify trends and create models to predict respiratory health issues based on air quality variables. This approach can help address the problem by identifying the specific variables or pollutants that contribute most to respiratory health issues, which can then be used to advise policies and interventions that can regulate these most dangerous pollutants.

Data

Annual AQI by County - Source: US Environmental Protection Agency (https://aqs.epa.gov/aqsweb/airdata/download_files.html) This dataset contains Air Quality Index values by county for 2023. The AQI is calculated each day for the Criteria Gases and PM10 and PM2.5. The variables are tracked in number of days each pollutant is observed, as well as number of days with good or poor AQI. It should be noted that different counties reported AQI different amounts of days, so there are some counties with more data than others.

Chronic Lower Respiratory Disease Mortality by State – Source: CDC National Center for Health Statistics (https://www.cdc.gov/nchs/pressroom/sosmap/lung_disease_mortality/lung_disease.htm) This dataset contains the rate of deaths per 100,000 total population from chronic lower respiratory disease for each state in the US. The variables in this dataset are State, rate of death, and number of deaths. One potential with this data is that when the number of deaths is small, the rate may be unreliable.

Asthma Hospitalization Rates by County – Source: California Department of Health Care Access and Information (<https://data.chhs.ca.gov/dataset/asthma-hospitalization-rates-by-county>) This dataset contains the number of hospitalizations for asthma by county in the state of California. The variables are county, year, age group, number of hospitalizations, and hospitalization rate. Some counties do not have hospitalization rates provided, but there is also a comment field providing an explanation when there is no rate.

Packages

tidyverse - For data manipulation and visualization ggplot2 – For creating plots dplyr – For data handling Metrics – for data modeling

Plots and tables

I will use histograms and scatter plots to show the different levels of pollutants and their relations to health outcomes. If possible I will also include heatmaps to provide geographical visualization of the data.

Questions for future steps

Can we find more datasets on different respiratory diseases for more geographic locations?

What models would be most effective at predicting respiratory health outcomes from the data provided?

What are other possible factors that could result in poor or missing data?

If trends are identified, how can air pollution successfully be mitigated in these locations?

Step 2

Importing cleaning, and manipulating the data

First we read the data from the csv files and store it in data frames.

```
aqi_data_2015 = read.csv("/Users/bryanemer/Downloads/annual_aqi_by_county_2015.csv")
aqi_data_2016 = read.csv("/Users/bryanemer/Downloads/annual_aqi_by_county_2016.csv")
aqi_data_2017 = read.csv("/Users/bryanemer/Downloads/annual_aqi_by_county_2017.csv")
aqi_data_2018 = read.csv("/Users/bryanemer/Downloads/annual_aqi_by_county_2018.csv")
aqi_data_2019 = read.csv("/Users/bryanemer/Downloads/annual_aqi_by_county_2019.csv")
aqi_data_2020 = read.csv("/Users/bryanemer/Downloads/annual_aqi_by_county_2020.csv")
asthma_data = read.csv("/Users/bryanemer/Downloads/asthma-hospitalization-rates-by-county-2015_2020.csv")
mortality_data = read.csv("/Users/bryanemer/Downloads/data-table.csv")
```

Next, we will join the AQI data from the different years to one table, then we will capitalize the column names to match the other tables. The mortality rate data also contains years outside of 2015 - 2020 that we need to remove.

```

aqi_data = bind_rows(aqi_data_2015, aqi_data_2016, aqi_data_2017, aqi_data_2018, aqi_data_2018, aqi_data_2018)
names(aqi_data) = base::toupper(names(aqi_data))
mortality_data = filter(mortality_data, YEAR >= 2015 & YEAR <= 2020)

```

Next, we will drop any missing values.

```

aqi_data = aqi_data %>% drop_na()
asthma_data = asthma_data %>% drop_na()
mortality_data = mortality_data %>% drop_na()

```

Then we will check for and remove any duplicate rows. Certain rows in the asthma data set have the “COUNTY” value as “California” which represents the entire state, so we will remove these rows. The data is also listed in different strata/population groups, so for now we only want the “Total Population” strata.

```

aqi_data = aqi_data %>% distinct()
asthma_data = asthma_data %>% distinct()
mortality_data = mortality_data %>% distinct()
asthma_data = asthma_data %>% filter(COUNTY != "California")
asthma_data = asthma_data %>% filter(STRATA == "Total population")

```

Next we will create a new data frame with AQI data just from California, and join that with the asthma data. Then we will drop rows with missing values again because the AQI data set has more counties listed than the asthma data set. We can also drop the COMMENT column as these values apply to counties where there is not a reported hospitalization rate. Finally, we will drop the NUMBER.OF.HOSPITALIZATIONS, STRATA, STRATA.NAME, and AGE.GROUP columns because we will not be using these values. We are concerned with the hospitalization rate, not the number of hospitalizations, and we are only looking at one strata.

```

ca_aqi_data = aqi_data %>% filter(STATE == "California")
ca_data = left_join(ca_aqi_data, asthma_data, by = c("COUNTY", "YEAR"))
ca_data = ca_data %>% drop_na()
ca_data$COMMENT = NULL
ca_data$NUMBER.OF.HOSPITALIZATIONS = NULL
ca_data$STRATA = NULL
ca_data$AGE.GROUP = NULL
ca_data$STRATA.NAME = NULL

```

Now we can display a short summary of the combined data, containing the columns and some entries of the data set as well as statistics regarding the columns -

```

glimpse(ca_data)

```

```

## Rows: 264
## Columns: 19
## $ STATE                <chr> "California", "California", "Calif~
## $ COUNTY               <chr> "Alameda", "Alpine", "Amador", "Bu~
## $ YEAR                 <int> 2015, 2015, 2015, 2015, 2015, 2015~
## $ DAYS.WITH.AQI        <int> 365, 31, 364, 365, 365, 365, 365, ~
## $ GOOD.DAYS            <int> 214, 1, 283, 170, 245, 258, 239, 6~
## $ MODERATE.DAYS        <int> 139, 14, 73, 182, 98, 99, 102, 203~
## $ UNHEALTHY.FOR.SENSITIVE.GROUPS.DAYS <int> 12, 13, 8, 12, 18, 8, 22, 78, 0, 0~

```

```
## $ UNHEALTHY.DAYS          <int> 0, 3, 0, 1, 2, 0, 2, 23, 0, 0, 3, ~
## $ VERY.UNHEALTHY.DAYS     <int> 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1~
## $ HAZARDOUS.DAYS         <int> 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0~
## $ MAX.AQI                 <int> 147, 170, 140, 152, 320, 147, 166,~
## $ X90TH.PERCENTILE.AQI    <int> 76, 149, 67, 84, 90, 67, 93, 136, ~
## $ MEDIAN.AQI              <int> 46, 102, 41, 52, 45, 44, 44, 80, 4~
## $ DAYS.CO                 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ DAYS.NO2                <int> 15, 0, 0, 0, 0, 7, 0, 0, 0, 0, 5, ~
## $ DAYS.OZONE              <int> 180, 0, 364, 237, 274, 213, 350, 2~
## $ DAYS.PM2.5              <int> 170, 31, 0, 127, 91, 145, 5, 133, ~
## $ DAYS.PM10               <int> 0, 0, 0, 1, 0, 0, 10, 5, 23, 6, 86~
## $ AGE.ADJUSTED.HOSPITALIZATION.RATE <dbl> 9.3, 0.0, 7.5, 6.7, 6.5, 8.3, 4.0,~
```

```
summary(ca_data)
```

```
##      STATE      COUNTY      YEAR      DAYS.WITH.AQI
## Length:264      Length:264      Min.    :2015      Min.    : 24.0
## Class :character Class :character 1st Qu.:2016      1st Qu.:365.0
## Mode  :character Mode  :character Median :2017      Median :365.0
##                                     Mean  :2017      Mean  :361.6
##                                     3rd Qu.:2019      3rd Qu.:366.0
##                                     Max.   :2020      Max.   :366.0
##      GOOD.DAYS  MODERATE.DAYS  UNHEALTHY.FOR.SENSITIVE.GROUPS.DAYS
## Min.    : 1      Min.    : 0.00      Min.    : 0.00
## 1st Qu.:147      1st Qu.: 65.75      1st Qu.: 3.75
## Median :232      Median :112.50      Median : 11.00
## Mean    :212      Mean    :117.57      Mean    : 23.76
## 3rd Qu.:290      3rd Qu.:171.25      3rd Qu.: 33.00
## Max.    :360      Max.    :257.00      Max.    :115.00
## UNHEALTHY.DAYS  VERY.UNHEALTHY.DAYS  HAZARDOUS.DAYS      MAX.AQI
## Min.    : 0.000      Min.    : 0.0000      Min.    :0.0000      Min.    : 50.0
## 1st Qu.: 0.000      1st Qu.: 0.0000      1st Qu.:0.0000      1st Qu.: 136.0
## Median : 2.000      Median : 0.0000      Median :0.0000      Median : 166.0
## Mean    : 7.314      Mean    : 0.8106      Mean    :0.1439      Mean    : 194.2
## 3rd Qu.: 9.000      3rd Qu.: 0.0000      3rd Qu.:0.0000      3rd Qu.: 200.2
## Max.    :69.000      Max.    :23.0000      Max.    :9.0000      Max.    :3852.0
## X90TH.PERCENTILE.AQI  MEDIAN.AQI      DAYS.CO      DAYS.NO2
## Min.    : 40.00      Min.    : 22.00      Min.    :0.00000      Min.    : 0.00
## 1st Qu.: 61.00      1st Qu.: 39.00      1st Qu.:0.00000      1st Qu.: 0.00
## Median : 80.00      Median : 44.00      Median :0.00000      Median : 0.00
## Mean    : 86.36      Mean    : 49.69      Mean    :0.07197      Mean    : 4.97
## 3rd Qu.:103.25      3rd Qu.: 56.00      3rd Qu.:0.00000      3rd Qu.: 3.00
## Max.    :185.00      Max.    :102.00      Max.    :3.00000      Max.    :68.00
##      DAYS.OZONE  DAYS.PM2.5  DAYS.PM10
## Min.    : 0      Min.    : 0.0      Min.    : 0.00
## 1st Qu.:191      1st Qu.: 81.5      1st Qu.: 0.00
## Median :223      Median :124.0      Median : 2.00
## Mean    :229      Mean    :116.4      Mean    : 11.25
## 3rd Qu.:259      3rd Qu.:157.0      3rd Qu.: 12.00
## Max.    :366      Max.    :343.0      Max.    :130.00
## AGE.ADJUSTED.HOSPITALIZATION.RATE
## Min.    : 0.000
## 1st Qu.: 2.975
## Median : 4.000
```

```
## Mean      : 4.179
## 3rd Qu.: 5.300
## Max.      :12.000
```

Now we need to create a new data frame containing AQI data by state instead of county because our death rate data is listed by State. We also need to convert the state names from abbreviations to full names in the mortality data so it can be joined with the AQI data.

```
mortality_data$STATE = state.name[match(mortality_data$STATE, state.abb)]
aqi_mort = left_join(aqi_data, mortality_data, by = c("STATE", "YEAR"))
summary(aqi_mort)
```

```
##      STATE          COUNTY          YEAR    DAYS.WITH.AQI
## Length:6145      Length:6145      Min.   :2015      Min.    : 2.0
## Class :character  Class :character  1st Qu.:2016      1st Qu.:288.0
## Mode  :character  Mode  :character  Median :2017      Median :362.0
##                                     Mean  :2017      Mean   :316.6
##                                     3rd Qu.:2019      3rd Qu.:365.0
##                                     Max.   :2020      Max.   :366.0
##
##      GOOD.DAYS      MODERATE.DAYS      UNHEALTHY.FOR.SENSITIVE.GROUPS.DAYS
## Min.   : 1.0      Min.   : 0.00      Min.   : 0.000
## 1st Qu.:219.0      1st Qu.: 20.00      1st Qu.: 0.000
## Median :280.0      Median : 42.00      Median : 1.000
## Mean   :259.1      Mean   : 53.53      Mean   : 3.096
## 3rd Qu.:318.0      3rd Qu.: 76.00      3rd Qu.: 3.000
## Max.   :365.0      Max.   :339.00      Max.   :115.000
##
##      UNHEALTHY.DAYS      VERY.UNHEALTHY.DAYS      HAZARDOUS.DAYS      MAX.AQI
## Min.   : 0.0000      Min.   : 0.0000      Min.   : 0.00000      Min.   : 11.0
## 1st Qu.: 0.0000      1st Qu.: 0.0000      1st Qu.: 0.00000      1st Qu.: 84.0
## Median : 0.0000      Median : 0.0000      Median : 0.00000      Median : 101.0
## Mean   : 0.7437      Mean   : 0.1056      Mean   : 0.06184      Mean   : 119.5
## 3rd Qu.: 0.0000      3rd Qu.: 0.0000      3rd Qu.: 0.00000      3rd Qu.: 133.0
## Max.   :69.0000      Max.   :74.0000      Max.   :26.00000      Max.   :6403.0
##
##      X90TH.PERCENTILE.AQI      MEDIAN.AQI      DAYS.CO      DAYS.NO2
## Min.   : 6.00      Min.   : 1.0      Min.   : 0.0000      Min.   : 0.000
## 1st Qu.: 48.00      1st Qu.: 33.0      1st Qu.: 0.0000      1st Qu.: 0.000
## Median : 54.00      Median : 37.0      Median : 0.0000      Median : 0.000
## Mean   : 56.07      Mean   : 35.8      Mean   : 0.6135      Mean   : 5.129
## 3rd Qu.: 61.00      3rd Qu.: 41.0      3rd Qu.: 0.0000      3rd Qu.: 0.000
## Max.   :215.00      Max.   :115.0      Max.   :328.0000      Max.   :365.000
##
##      DAYS.OZONE      DAYS.PM2.5      DAYS.PM10      RATE
## Min.   : 0.0      Min.   : 0.0      Min.   : 0.00      Min.   :17.30
## 1st Qu.: 96.0      1st Qu.: 2.0      1st Qu.: 0.00      1st Qu.:35.10
## Median :204.0      Median :107.0      Median : 0.00      Median :41.60
## Mean   :181.6      Mean   :117.1      Mean   :12.17      Mean   :42.76
## 3rd Qu.:261.0      3rd Qu.:180.0      3rd Qu.: 0.00      3rd Qu.:48.50
## Max.   :366.0      Max.   :366.0      Max.   :366.00      Max.   :66.70
##                                     NA's    :91
##      DEATHS      URL
```

```
## Length:6145      Length:6145
## Class :character  Class :character
## Mode :character  Mode :character
##
##
##
##
```

Questions for Future Steps

In order to fully utilize this data, we need to take the averages of each column by state and create a new data frame that lists each state's AQI data along with the mortality data. I am not currently sure how to effectively do this, so further research is required. However, we now have a useful data set containing the AQI data for each county in California along with the asthma related hospitalization rates.

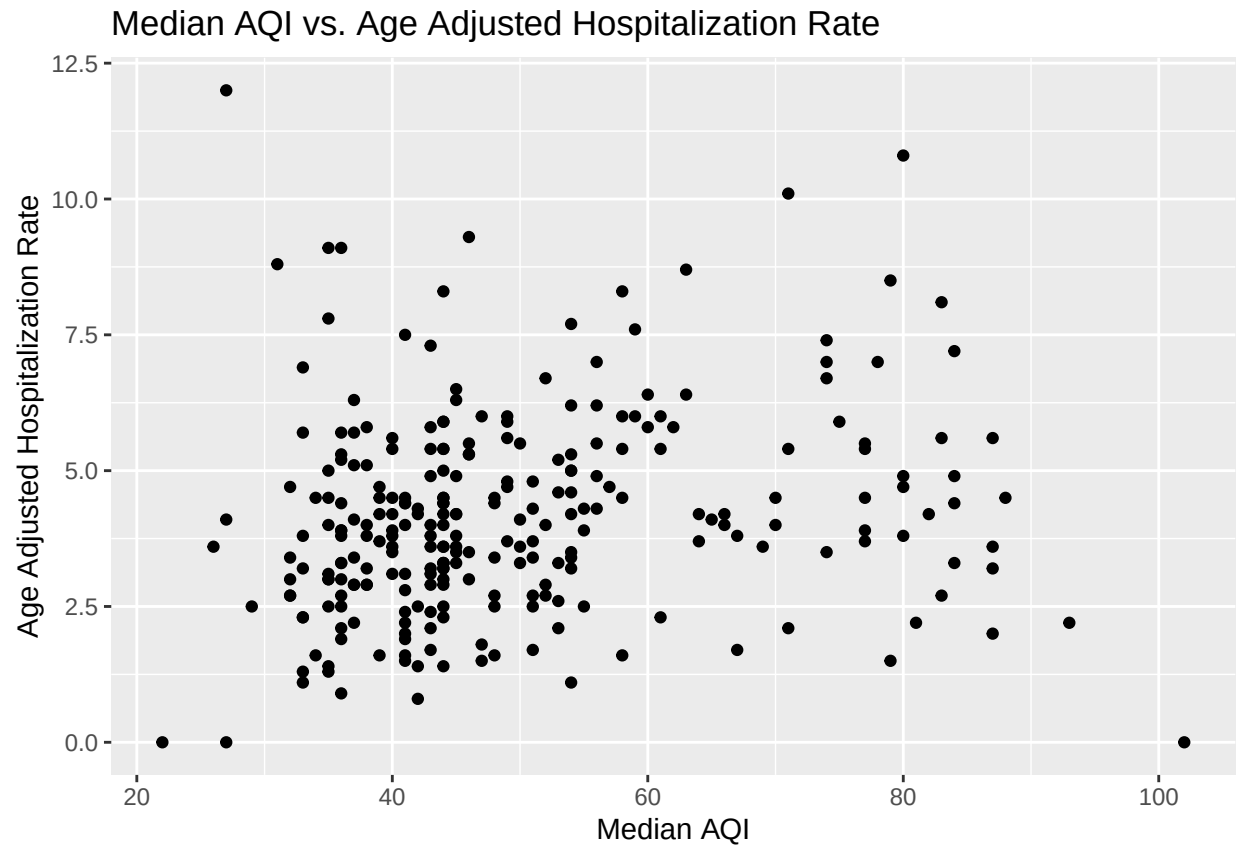
What are Different Ways you Could Look at This Data?

We could further divide the data by year, or look specifically at which pollutants are related most with asthma hospitalization rates. We could also look at how the number of good AQI days vs. unhealthy AQI days related to the hospitalization rate as opposed to just looking at the median AQI.

How could you summarize your data to answer key questions?

We will show different plots to demonstrate the relationship between median AQI and Hospitalization rate, along with plots showing the relationship between different pollutants and hospitalization rate. We can also use Pearson's correlation test to find the correlation values between these variables and compare the results.

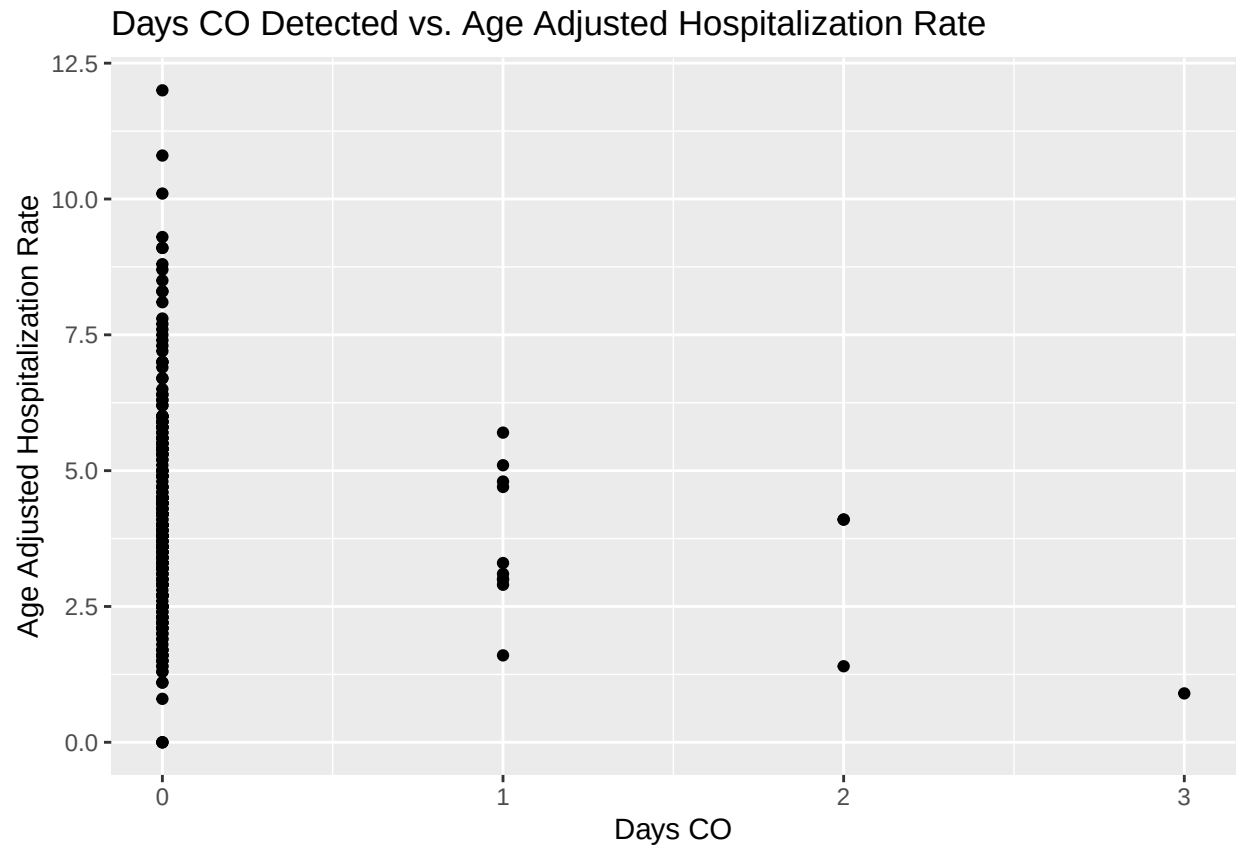
```
ggplot(data = ca_data, aes(x = MEDIAN.AQI, y = AGE.ADJUSTED.HOSPITALIZATION.RATE)) +
  geom_point() + labs(title = "Median AQI vs. Age Adjusted Hospitalization Rate", x = "Median AQI", y =
```



```
print(paste("Correlation between Median AQI and Hospitalization Rate: ", round(cor(ca_data$MEDIAN.AQI, AGE.ADJUSTED.HOSPITALIZATION.RATE), 2)))
```

```
## [1] "Correlation between Median AQI and Hospitalization Rate: 0.174"
```

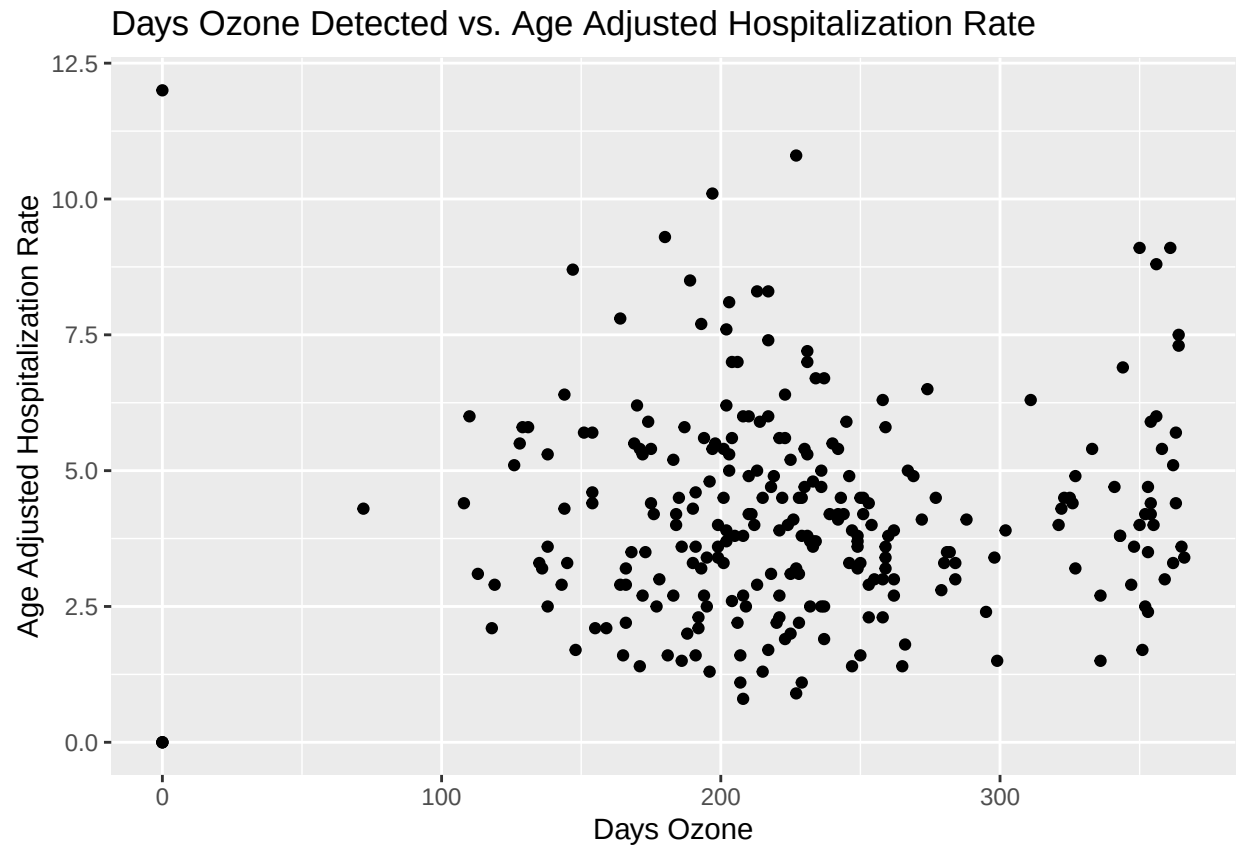
```
ggplot(data = ca_data, aes(x = DAYS.CO, y = AGE.ADJUSTED.HOSPITALIZATION.RATE)) +  
  geom_point() + labs(title = "Days CO Detected vs. Age Adjusted Hospitalization Rate", x = "Days CO", y = "Age Adjusted Hospitalization Rate")
```



```
print(paste("Correlation between Days CO Detected and Hospitalization Rate: ", round(cor(ca_data$DAYS.CO,
```

```
## [1] "Correlation between Days CO Detected and Hospitalization Rate: -0.121"
```

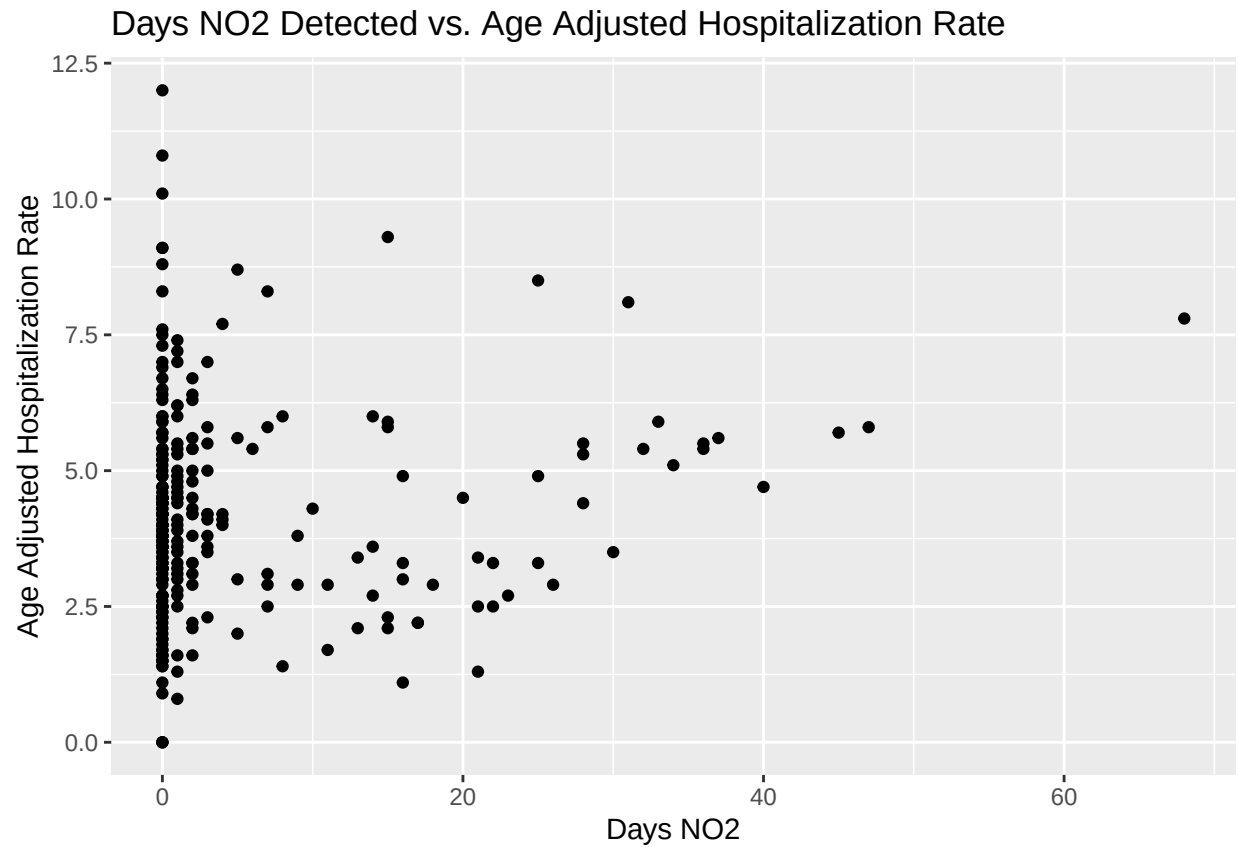
```
ggplot(data = ca_data, aes(x = DAYS.OZONE, y = AGE.ADJUSTED.HOSPITALIZATION.RATE)) +  
  geom_point() + labs(title = "Days Ozone Detected vs. Age Adjusted Hospitalization Rate", x = "Days Oz
```

```
print(paste("Correlation between Days Ozone Detected and Hospitalization Rate: ", round(cor(ca_data$DAY
```

```
## [1] "Correlation between Days Ozone Detected and Hospitalization Rate: 0.064"
```

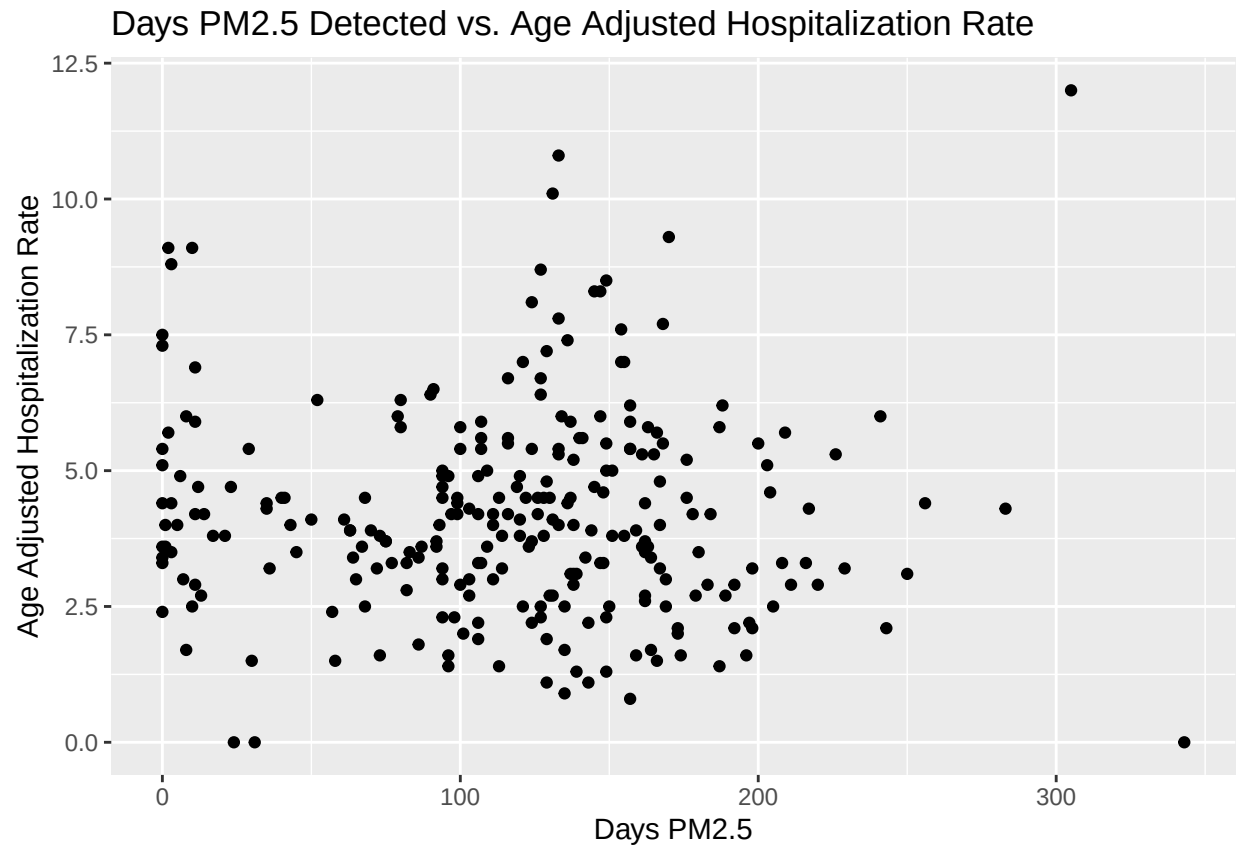
```
ggplot(data = ca_data, aes(x = DAYS.NO2, y = AGE.ADJUSTED.HOSPITALIZATION.RATE)) +  
  geom_point() + labs(title = "Days NO2 Detected vs. Age Adjusted Hospitalization Rate", x = "Days NO2"
```



```
print(paste("Correlation between Days NO2 Detected and Hospitalization Rate: ", round(cor(ca_data$DAYS.NO2, ca_data$AGE.ADJUSTED.HOSPITALIZATION.RATE), 2)))
```

```
## [1] "Correlation between Days NO2 Detected and Hospitalization Rate: 0.119"
```

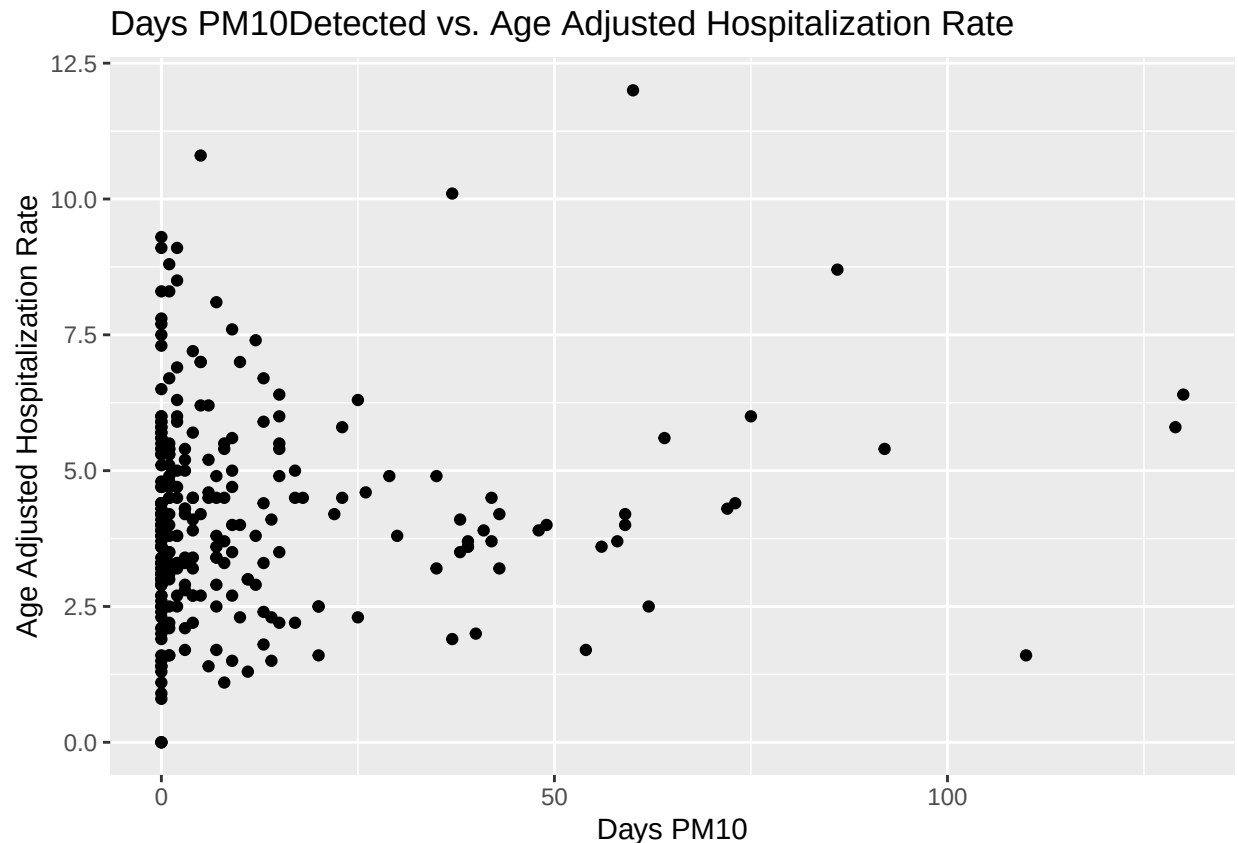
```
ggplot(data = ca_data, aes(x = DAYS.PM2.5, y = AGE.ADJUSTED.HOSPITALIZATION.RATE)) +  
  geom_point() + labs(title = "Days PM2.5 Detected vs. Age Adjusted Hospitalization Rate", x = "Days PM2.5 Detected")
```



```
print(paste("Correlation between Days PM2.5 Detected and Hospitalization Rate: ", round(cor(ca_data$DAYSPM25, ca_data$AGEADJUSTEDHOSPITALIZATIONRATE), 2)))
```

```
## [1] "Correlation between Days PM2.5 Detected and Hospitalization Rate: -0.027"
```

```
ggplot(data = ca_data, aes(x = DAYS.PM10, y = AGE.ADJUSTED.HOSPITALIZATION.RATE)) +  
  geom_point() + labs(title = "Days PM10Detected vs. Age Adjusted Hospitalization Rate", x = "Days PM10")
```



```
print(paste("Correlation between Days PM10 Detected and Hospitalization Rate: ", round(cor(ca_data$DAYS
## [1] "Correlation between Days PM10 Detected and Hospitalization Rate:  0.092"
```

By comparing these results, we can see that median AQI is overall the most correlated with asthma hospitalization rates. Of the recorded pollutants, NO₂ has the highest correlation with hospitalization rates. PM₁₀ also appears to be slightly correlated with hospitalization rates. Using this data we can determine that if we want to reduce asthma hospitalization rates in a given region, we should focus our efforts on limiting the amount of NO₂ that enters the atmosphere.

Questions for further steps

One question we could ask is how can we find more data on air pollution to improve our ability to predict hospitalization rates? There is likely to be other pollution data sets we can use. Another question would be how can we use machine learning models to predict hospitalization rates? We could possibly fit a linear regression model to this data. How can we fine tune these models even further? Another question would be knowing that NO₂ is a dangerous pollutant, what are the leading causes of NO₂ being emitted and how can we prevent it?

Step 3

The core problem addressed in this project was the impact of urban air pollution on health, specifically respiratory health outcomes. As our cities grow urban air pollution will continue to be an important issue, and we hope to help understand how air pollution contributes to respiratory health outcomes.

How We Addressed the Problem

To address this problem, we utilized various data sets for government organizations tracking air quality measurements and respiratory health outcomes in the United States. First, we had to perform data cleaning. The data was joined together to create a new data set containing the air quality measurements and asthma hospital admission rates for the state of California. Analyzing this new data set allowed us to observe the correlation between the different pollutants in the air quality data and the asthma hospital admission rates. Although a predictive model was not built, a regression model could potentially be used to predict asthma hospitalization rates based on pollutant levels.

Analysis

The analysis of the data set revealed some key insights into the relationship between air pollutants and asthma hospitalization rates. We found that the median AQI for a given county is the best predictor for asthma hospitalization in a given county. When breaking down the specific pollutants, we found that NO₂ had the highest correlation with asthma hospitalization rates, followed by PM₁₀. The data shows us that these pollutants tend to have the highest impact on respiratory health issues.

Implications

The implications of this analysis are significant for both the public and policymakers. If we wish to craft policy to address air pollution and reduce asthma hospitalization rates, it would make sense to focus these efforts on pollutants like NO₂ and PM₁₀ given the results of our data analysis. There should be efforts to determine what most causes the emission of these pollutants and then craft policy to reduce these emissions. Individuals can also use this information to make informed decisions about their health and take precautionary measures when there are high concentrations of these pollutants in the air. Public health campaigns could be used to inform the public about the dangers of these pollutants.

Limitations

One limitation was the availability of data. We were only able to locate asthma hospital admission data for California, which limits the scope of our analysis. Finding more data on this for the rest of the country, along with information about other respiratory health issues would increase the accuracy of our predictions. This also limits the scope of our findings, as there may be different environmental conditions in different urban areas. We also cannot conclude from this analysis that these pollutants necessarily cause asthma hospitalization, only that they are correlated.

Conclusion

This project demonstrated the significant relationship between air pollution and respiratory health, providing insights that can be used to inform health decisions both for the public and policymakers. While there are some limitations to this analysis, our findings show the importance of monitoring and regulating air quality in urban areas. We can build on this research by creating predictive models for asthma hospitalization and further exploring the relationships between air pollutants and respiratory health outcomes. This is crucial research as air pollution will continue to be a major factor in respiratory health outcomes as urban areas grow.